



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP2523 DATABASE (WBL)

NAME: CHOH JING YI (A23CS0296)

**SECTION 02
ASSIGNMENT 1**

LECTURER: DR. SEAH CHOON SEN

QUESTION 1

One of the application areas of database systems is e-commerce such as online shopping.

What are the data or information stored in the database system of an online shopping website? Give FOUR (4) of them.

Data	Customer ID	Product ID	Order ID	Transaction ID
Information	Customer details (name, address, contact information)	Product details (name, description, price)	Order details (order status, delivery date)	Payment information (total amount, payment method, discount applied)

QUESTION 2

Answer the questions below based on the following relational schema:

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SCIENTIST (Sc_ID, Sc_Name, Sc_Phone, Sc_Add)
EXPERIMENT (Exp_NO, Ex_Name, Proj_Code)
PROJECT (Proj_Code, Proj_Name, Proj_StartDate)
SCIENTIST_EXPERIMENT (Sc_ID, Exp_NO, Date, Time)
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- a) For each table, identify the primary key and foreign key. If a table does not have a primary key or foreign key, write – in the space provided. (4M)

TABLE	PRIMARY KEY	FOREIGN KEY
SCIENTIST	Sc_ID	-
EXPERIMENT	Exp_NO	Proj_Code
PROJECT	Proj_Code	-
SCIENTIST_EXPERIMENT	Sc_ID, Exp_NO	Sc_ID ,Exp_NO

- b) Does the table exhibit entity integrity? Answer ‘Yes’ or ‘No’ and then explain your answer. (4M)

TABLE	ENTITY INTEGRITY	EXPLANATION
SCIENTIST	Yes	The primary key Sc_ID ensures entity uniqueness.
EXPERIMENT	Yes	The primary key Exp_NO ensures entity uniqueness.
PROJECT	Yes	The primary key Proj_Code ensures entity uniqueness.

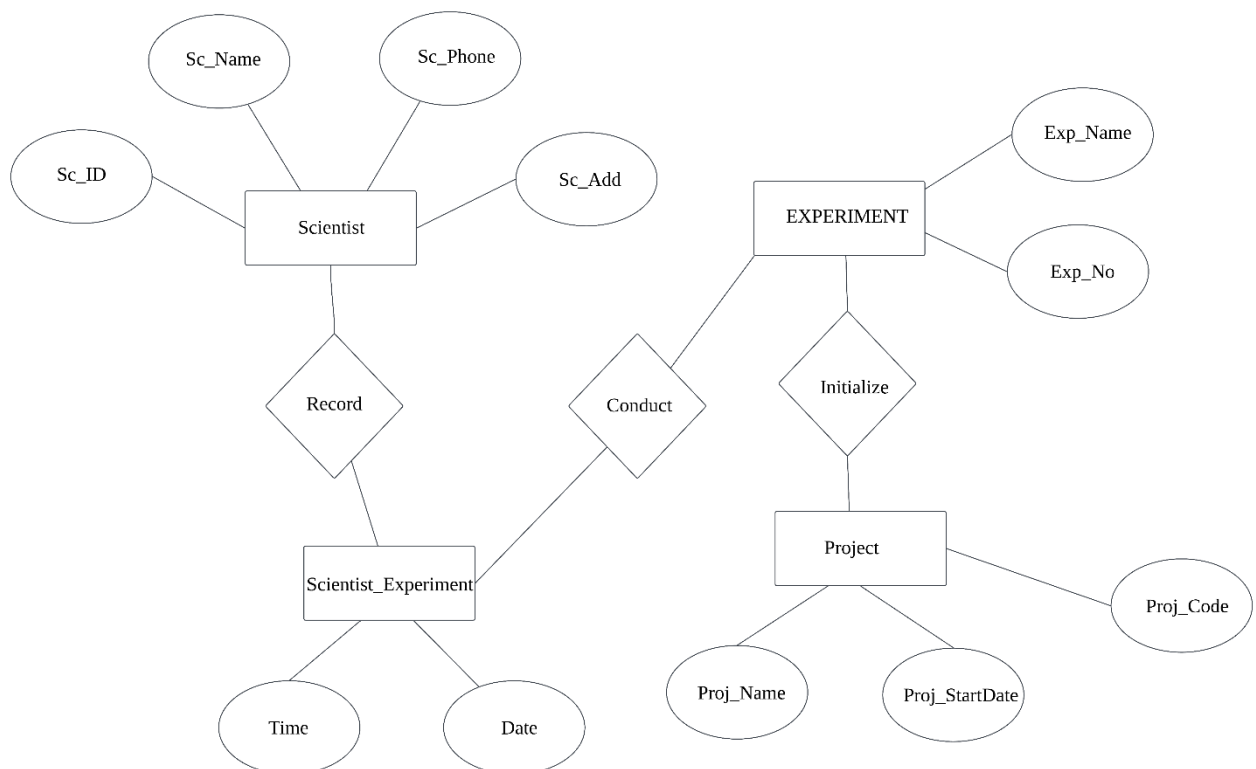
ASSIGNMENT 1 SECP2523 DATABASES

SCIENTIST_EXPERIMENT	Yes	The composite primary key (Sc_ID, Exp_NO) ensures uniqueness for the relationship.
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- c) Does the table exhibit referential integrity? Answer ‘Yes’ or ‘No’ and then explain your answer. (4M)

TABLE	REFERENTIAL INTEGRITY	EXPLANATION
SCIENTIST	No	No foreign key exists in this table, so referential integrity is not applicable.
EXPERIMENT	Yes	Proj_Code is a foreign key, and it must exist in the Project table for referential integrity to be maintained.
PROJECT	No	No foreign key exists in this table, so referential integrity is not applicable.
SCIENTIST_EXPERIMENT	Yes	Sc_ID and Exp_NO are foreign keys, and they must exist in the SCIENTIST and EXPERIMENT tables, respectively, to maintain referential integrity.

- d) Draw ERD using Chen notation based on the given relational schema. (4M)



QUESTION 3

Read the following and complete the task listed.

Consider the following about Luxury Records:

Here is the information that you gather:

- Luxury has decided to store information about musicians who perform on its albums (as well as other company data) in a database.
- Each musician that records at Luxury has an SSN, a name, an address, and a phone number.
- Poorly paid musicians often share the same address, and no address has more than one phone.
- Each instrument used in songs recorded at Luxury has a unique identification number, a name (e.g guitar, synthesizer, and a flute) and a musical key (e.g., C, B-flat, E-flat).
- Each album recorded on the Luxury label has a unique identification number, a title, a copyright date, a format (e.g., CD or MC) and an album identifier.
- Each song recorded at Luxury has a title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each song album has a number of songs on it, but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform a number of songs.
- Each album has exactly one musician who acts as its producer.
- A musician may produce several albums, of course.

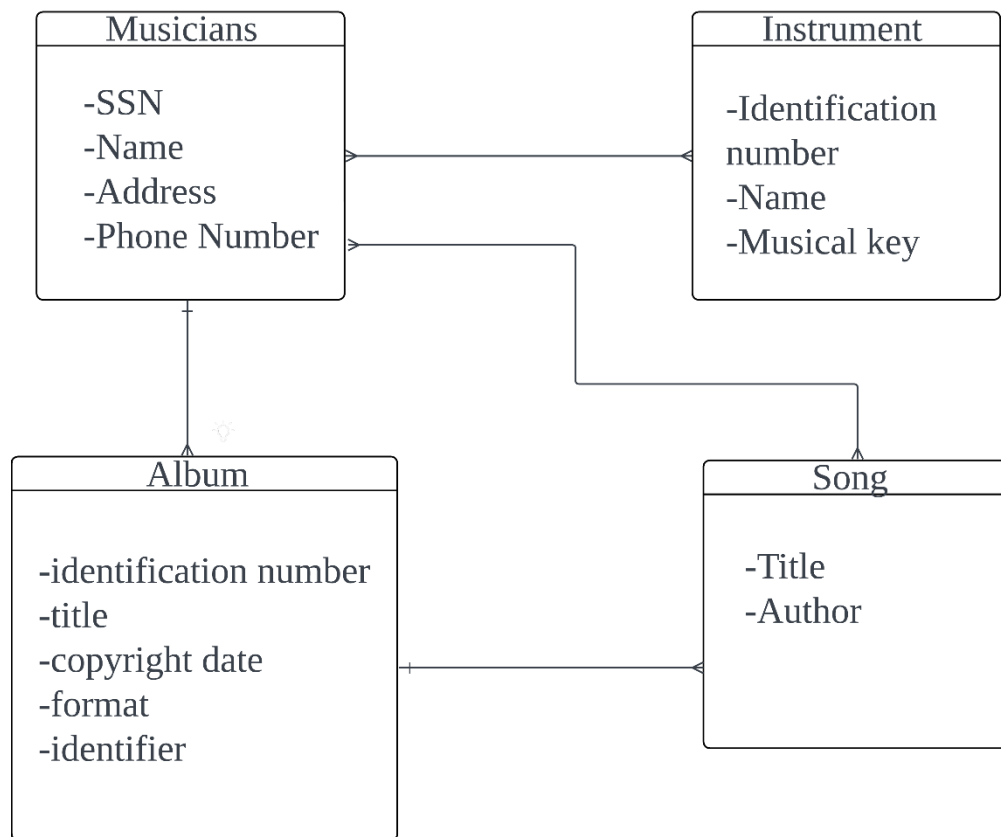
Task:

1. List each Entity that is identified in Luxury Records System.
-musician
-instrument
-album
-song

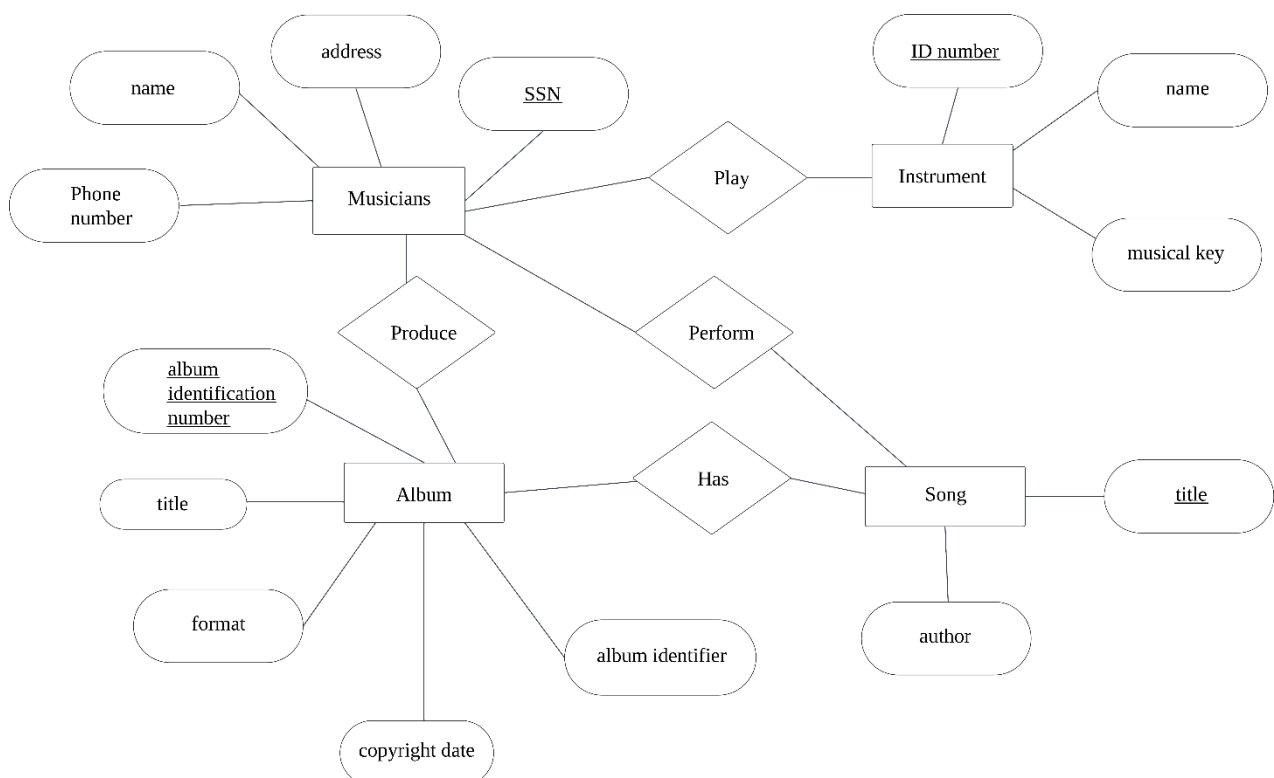
2. For each entity list the attributes.

Musician	SSN, name, address, phone number
Instrument	ID number, name, musical key
Album	Identification number, title, copyright date, format, album identifier
Song	Title, author

3. Draw a logical model to represent each entity and the relationships between them.



4. Draw an Entity Relationship Diagram (ERD) that captures the entire system.



QUESTION 4

Give an example of each of the three types of relationships.

i) One-to-many relationship

-Musician to Albums

A single musician can produce multiple albums, but each album is produced by only one musician.

ii) Many-to-many relationship

-Musicians to instruments.

Each musician may play several instruments, and a given instrument may be played by several musicians.

iii) One-to-one relationship

-Musician to Phone Number

Each musician has a unique phone number, and no two musicians can share the same phone number.

iv) Many-to-one relationship

-Songs to Album

Each song is part of one album, but an album contains multiple songs.

QUESTION 5

CLIENT (Client_Number, Client_Name, Street, City, State, Postal_Code, Amount_Paid, Current_Due, Recruiter_Number)

RECRUITER (Recruiter_Number, Last_Name, First_Name, Street, City, State, Postal_Code, Rate, Commission)

a) What is the relationship between Recruiter and Client?

One to many

b) What is the primary key for the Client table and Recruiter table?

Client table: The primary key is Client_Number

Recruiter table: The primary key is Recruiter_Number

c) Define foreign key? Identify foreign key in the Client table.

A foreign key (FK) is an attribute of an entity which serve for another entity.

The foreign key in the Client table is Recruiter_Number.

QUESTION 6

Draw the Chen & Crow's Foot notation based on the following statements:

- i) "A customer can make many payments, but each payment is made by only one customer."



- ii) "A medical doctor may treat many patients. Some patients may be treated by more than one medical doctor. A patient may have or may not have insurance plan."



QUESTION 7

Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

STUDENT(SSN, Name, Major, Bdate)

COURSE(Course#, Cname, Dept)

ENROLL(SSN, Course#, Quarter, Grade)

BOOK_ADOPTION(Course#, Quarter, Book_ISBN)

TEXT(Book ISBN, Book_Title, Publisher, Author)

Draw a relational schema diagram specifying the foreign keys for this schema.

